OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	Latex Report Enhancement
Project Location	Mumbai(Bombay) (Santa Cruz), Maharashtra
Author / Designer	Dev Agrawal
Reviewer	
Organization	
Client Name and Organization	IIT Bombay
Job Number	
Date	2026-08-21
Report Version	1

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Executive Summary

This section provides a concise summary of the bridge design, key inputs, governing loads, and final design outcomes.

Project Overview

Bridge Type	Highway Bridge
Design Standard	IRC 5, IRC 6, IRC 22, IRC 24, IS 800
Span	30 m
Carriageway Width	7.5 m
No. of Girders	4
Girder Spacing	3.0375
Deck Thickness	250.0
Overall Design Status	Fail (Fatigue Normal Stress, Int.Stiff: Leg H_ limit, Fatigue Shear Stress)
Governing Check	Fatigue Normal Stress
Overall Utilization Ratio (max)	2.09 (Girder)

Figure 1 – Overall Bridge Plan

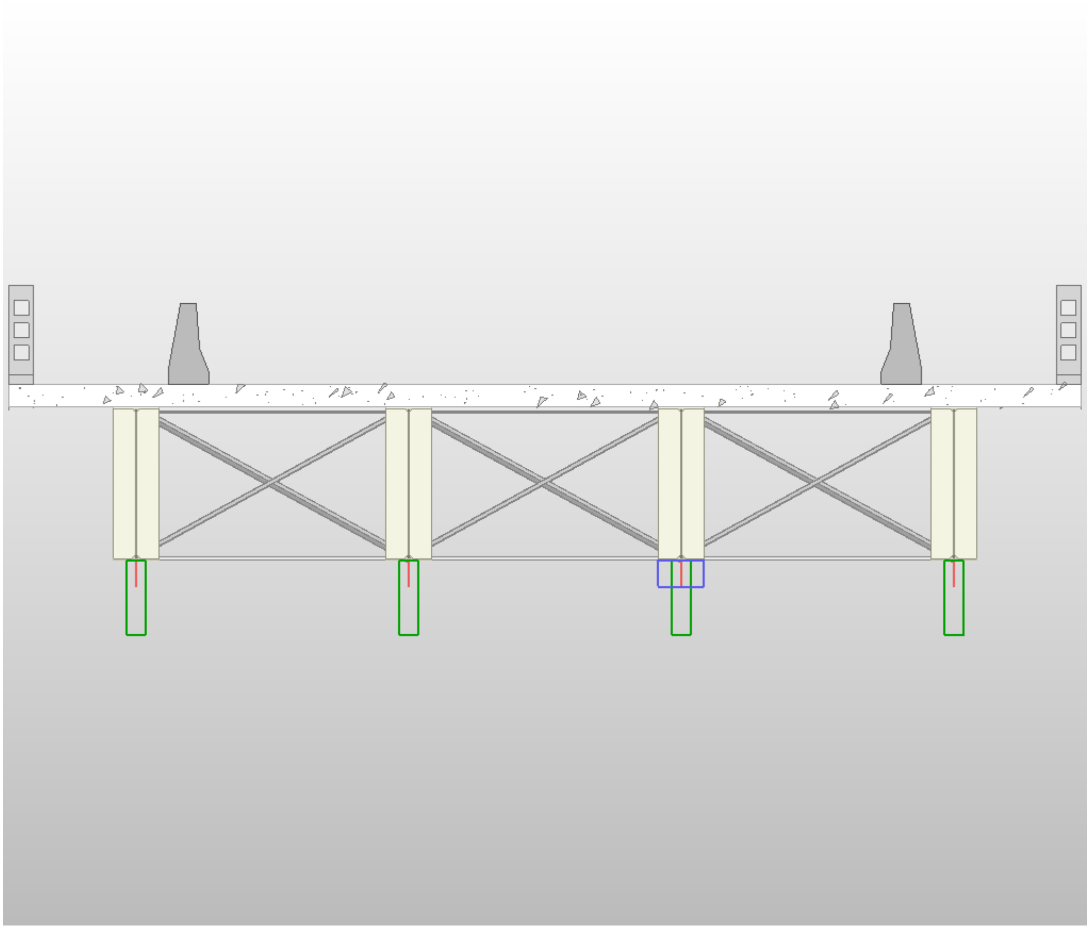


Figure 2 – Typical Cross-Section (with girder, deck, barriers, footpath)

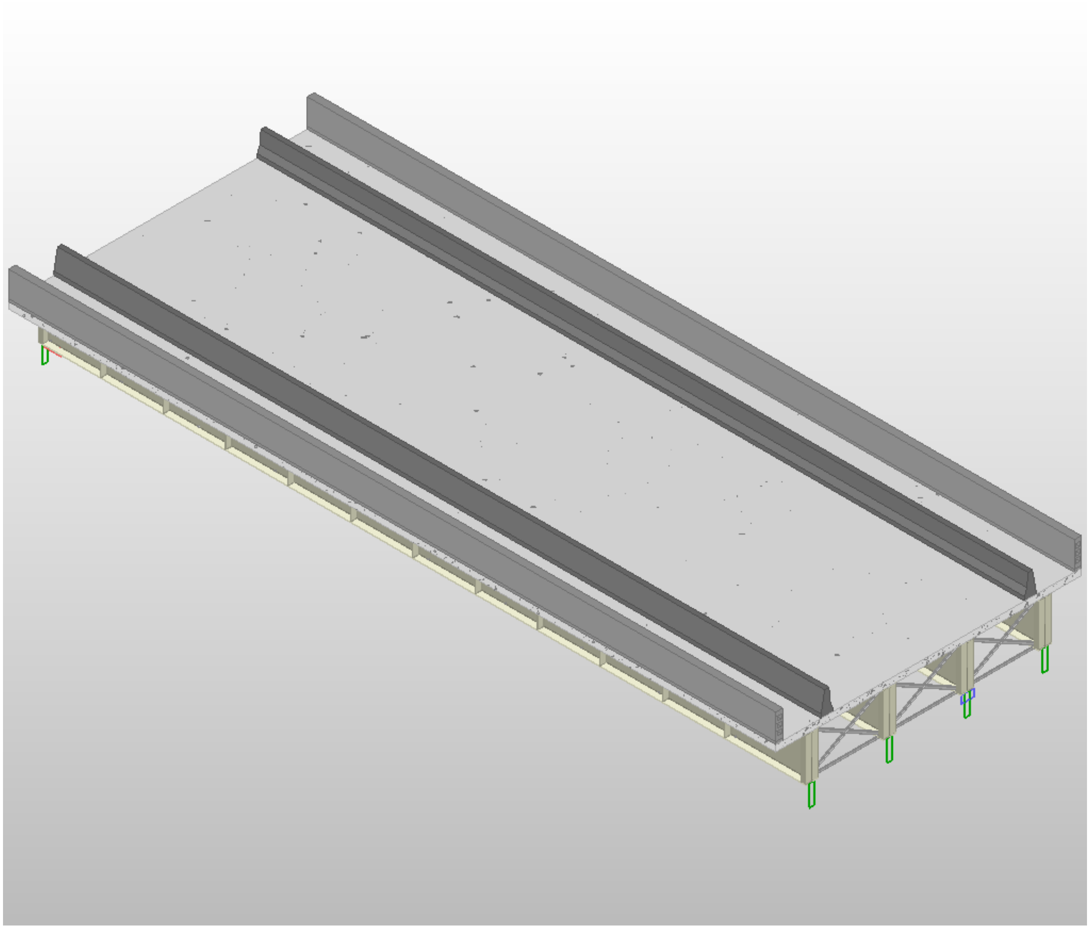


Figure 3 – 3D View of Bridge Superstructure

Table 1 – Final Bridge Geometry (after optimization)

	G1	**G2**	**G3**	**G4**
Member ID	G1M1	G2M1	G3M1	G4M1
Section Designation	1670 x 510 x 22 x 510 x 22	1670 x 510 x 22 x 510 x 22	1670 x 510 x 22 x 510 x 22	1670 x 510 x 22 x 510 x 22
Governing Check	Fatigue Normal Stress	Fatigue Normal Stress	Fatigue Normal Stress	Fatigue Normal Stress
Utilization Ratio	2.09 (Girder)	2.09 (Girder)	2.09 (Girder)	2.09 (Girder)

Note: Utilization ratio (UR) = demand / capacity. A value < 1.0 *indicates a passing check.*

Key Design Outcomes Summary

Girder design pass
Cross bracing design pass
End Diaphragm design pass
Deck design pass

Design Assumptions and Limitations

- Additional inputs not provided by the user were assumed by software per IRC/IS code defaults or practical consideration.
- Grillage analysis was performed using OSPGrillage assuming simply supported I-girders.
- Substructure and foundation design are not included in this report.
- Splice connections and bearings are not designed in this version.

1 Project Information

This section records all project metadata as entered by the designer.

1.1 Project and Design Team Details

Project Name	Latex Report Enhancement
Project Location	Mumbai(Bombay) (Santa Cruz), Maharashtra
Designer	Dev Agrawal
Reviewer	
Organization	
Client	IIT Bombay
Software Version	OsdagBridge

1.2 Applicable Codes and Standards

- Indian Roads Congress (IRC) 5: General Features of Design
- Indian Roads Congress (IRC) 6: Loads and Load Combinations
- Indian Roads Congress (IRC) 22: Composite Construction (Limit State Design)
- Indian Roads Congress (IRC) 24: Steel Road Bridges (Limit State Method)
- Indian Roads Congress (IRC) 112: Concrete Road Bridges (deck design)
- Indian Roads Congress Special Publication (IRC SP) 114: Seismic Design of Road Bridges
- Indian Standard (IS) 800: General Construction in Steel
- Indian Standard (IS) 2062: Hot Rolled Structural Steel Specification
- Indian Standard (IS) 6006: Steel Bearings

2 Input Parameters

This section documents all inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

Note: These inputs are mandatory and were provided by the user.

Table 2.1: Project Location

parameter	value
Project Location	Mumbai(Bombay) (Santa Cruz), Maharashtra
Latitude / Longitude	19.076, 72.8777
Seismic Zone (IRC 6)	III
Basic Wind Speed (IRC 6)	44 m/s
Shade Temp. Max / Min (IRC 6)	42.2 °C / 7.4 °C

Table 2.2: Bridge Geometry

parameter	value
Type of Structure	Highway Bridge
Span (m)	30 m
Carriageway Width (m)	7.5 m
Include Median	No
Footpath	Both Sides
Skew Angle (degrees)	0° (IRC 24 Cl. 504.8 limit: ±15°)

Table 2.3: Material Selection

parameter	value
Girder Steel Grade (IS 2062)	E 350A
Cross Bracing Steel Grade	E 350A
End Diaphragm Steel Grade	E 350A
Concrete Deck Grade (IRC 22)	M40

** Software default value*

2.2 Additional Inputs

Where the user has modified additional inputs, those values are reported here. Where no modification was made, the software default is shown.

Table 2.4: Typical Section Details

parameter	value
Overall Bridge Width (m)	12.149999999999999
No. of Girders	4
Girder Spacing (m)	3.0375 m
Deck Overhang Width (m)	1.5187 m
Deck Thickness (mm)	250.0 mm
Footpath Width (m)	1.5 m (IRC 5 Cl. 104.3.6 min: 1.5 m)
No. of Traffic Lanes	(per IRC 5 Cl. 104.3.1)

Table 2.5: Components Details

parameter	value
Crash Barrier Type	IRC 5 - RCC Crash Barrier
Crash Barrier Load (kN/m)	6.81
Railing Type	IRC 5 - RCC Railing
Railing Load (kN/m)	6.806
Wearing Course Material	Concrete
Wearing Course Thickness (mm)	50.0 mm

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Optimized	Welded	Girder Symmetric
G2	G2M1	Optimized	Welded	Girder Symmetric
G3	G3M1	Optimized	Welded	Girder Symmetric
G4	G4M1	Optimized	Welded	Girder Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange (b_{tf}, t_{tf}) mm	Bottom Flange (b_{bf}, t_{bf}) mm
G1	1.67 mm	0.01 mm	0.51 mm, 0.022 mm	0.51 mm, 0.022 mm
G2	1.67 mm	0.01 mm	0.51 mm, 0.022 mm	0.51 mm, 0.022 mm
G3	1.67 mm	0.01 mm	0.51 mm, 0.022 mm	0.51 mm, 0.022 mm
G4	1.67 mm	0.01 mm	0.51 mm, 0.022 mm	0.51 mm, 0.022 mm

Table 2.8: Girder Restraint and Stiffener Details

Girder	**Torsional / Warping Restraint**	**Web Philosophy**	**Intermediate Stiffeners**	**Longitudinal Stiffeners**	**Bearing Stiffener**
G1	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G2	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G3	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G4	Fully Restrained, Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G1 to G2	B1M1 to B1M1	X-Bracing	IS 100 x 100 x 10	15.0 m
G2 to G3	B2M1 to B2M1	X-Bracing	IS 100 x 100 x 10	15.0 m
G3 to G4	B3M1 to B3M1	X-Bracing	IS 100 x 100 x 10	15.0 m

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Bracing	Bracing Section
G1 to G2	E1M1 to E1M2	Cross Bracing	IS 100 x 100 x 10
G2 to G3	E2M1 to E2M2	Cross Bracing	IS 100 x 100 x 10
G3 to G4	E3M1 to E3M2	Cross Bracing	IS 100 x 100 x 10

Table 2.11: Shear Connector Details

parameter	value
Stud Diameter (mm)	20.0 mm
Stud Height (mm)	100.0 mm
Stud f_y (MPa)	385.0 MPa
Stud f_u (MPa)	495.0 MPa
No. of Studs per Section	2

Note: All values are per IRC 22 Table 1 unless user-modified.

Table 2.12: Partial Safety Factors

parameter	**value**
γ_{M0} (Yielding / Buckling)	1.1
γ_{M1} (Ultimate Stress)	1.25
γ_{C} (Concrete, Basic)	1.5
γ_{s} (Reinforcement)	1.15
γ_{v} (Shear Connectors)	1.25
γ_{fft} (Fatigue Load)	1.0
γ_{Mft} (Fatigue Strength)	1.35

3 Loads and Load Combinations

This section summarizes all loads applied to the bridge and the load combinations considered for analysis and design.

Table 3.1: Dead Load – Self Weight

parameter	value
Steel Self-Weight Applied	76.4 kN/m^3
Concrete Deck Weight	24.5 kN/m^3
Self-Weight Factor	1.0

Table 3.2: Dead Load for Surfacing (DW)

parameter	value
Wearing Course Load	Concrete x 50.0
Additional SIDL (Crash Barrier)	6.81 kN/m per barrier
Railing Load	6.806 kN/m*

Table 3.3: Vehicle Live Loads (LL)

Vehicle	Total Load (kN)	Impact Factor	Braking Considered?	Braking Value (kN)	Eccentricity (m)
Class A	543474.00	1.207	No	–	–
Class 70R (Wheeled)	981000.00	See IRC 6	No	–	–
Class 70R (Tracked)	N/A	See IRC 6	No	–	–
Class AA (Wheeled)	N/A	N/A	No	–	–
Class AA (Tracked)	N/A	N/A	No	–	–
Class SV	3776.85	N/A	Yes	N/A	1.2
Class 70R (Bogie)	N/A	N/A	No	–	–
Class Fatigue	392400.00	N/A	No	–	–

* *Software default value*

Table 3.4: Footpath Live Load

Parameter	Unit	Value
Footpath Live Load	4.905 kN/m^2 (IRC 6 Cl. 206.1)	

Table 3.5: Wind Load (WL) — per IRC 6

parameter	value
Basic Wind Speed, Vb	44 m/s [from Project Location]
Terrain Type	Plain Terrain
Average Exposed Height, H (m)	10 m
Hourly Mean Wind Speed, Vz	
Hourly Wind Pressure, Pz	
Transverse Wind Force	
Longitudinal Wind Force	
Vertical Wind Force	

Table 3.6: Earthquake Load (EL) — per IRC 6

parameter	**value**
Seismic Zone	III [from Project Location]
Zone Factor, Z	0.16
Importance Factor, I	1.0
Type of Soil	Type I – Rocky or Hard
Sa/g	2.8
Horizontal Seismic Coefficient, Ah	0.224
Vertical Seismic Coefficient, Av	0.1493
Horizontal Seismic Force (longitudinal)	kN
Horizontal Seismic Force (transverse)	kN

Table 3.7: Temperature Load (TL) — per IRC 6

parameter	**value**
Maximum Shade Temperature	42.2 °C
Minimum Shade Temperature	7.4 °C
Effective Bridge Temp. Range	-2.60 to 57.20 °C
Temperature Rise / Fall for Design	+29.90 °C / −29.90 °C

Table 3.8: Load Combinations

Combination ID	Load Cases
ULS-01	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.50) + WL(0.90) + TL(0.90)
ULS-02	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(1.50) + TL(0.90)
ULS-03	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(1.15) + WL(0.90) + TL(1.50)
ULS-04	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + VC(1.00)
ULS-05	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + BI(1.00)
ULS-06	DL(1.00/1.00) + SIDL(1.00/1.00) + LL(0.75) + TL(0.50) + FB(1.00)
ULS-07	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(1.50)
ULS-08	DL(1.35/1.00) + SIDL(1.75/1.00) + LL(0.20) + TL(0.50) + EQ(0.75)
SLS-01	DL(1.00) + SIDL(1.20/1.00) + LL(1.00) + WL(0.60) + TL(0.60)
SLS-02	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(1.00) + TL(0.60)
SLS-03	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.60) + TL(1.00)
SLS-04	DL(1.00) + SIDL(1.20/1.00) + LL(0.75) + WL(0.50) + TL(0.50)
SLS-05	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.60) + TL(0.50)
SLS-06	DL(1.00) + SIDL(1.20/1.00) + LL(0.20) + WL(0.50) + TL(0.60)
SLS-07	DL(1.00) + SIDL(1.20/1.00) + LL(0.00) + WL(0.00) + TL(0.50)

Note: All IRC 6 load combinations are auto-generated by OsdagBridge. User-defined custom combinations, if any, are appended.

4 Analysis Results

A grillage model was used for structural analysis. The deck is idealized as a grid of elastic beam elements — longitudinal members represent the composite steel girders with effective slab, and transverse members represent the slab or cross frames. This section summarizes the critical output from that analysis.

Table 4.1: Summary of Maximum Demands

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SLS_RARE_2: 1.0DL + 1.2DW + 0.75LL + 1.0WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
1.5 EQ (a)	146.104	G1	15.000	27.761	G1	0.000	89.742	-89.742
BASIC_3: 1.35DL + 1.75DW + 1.15LL + 0.9WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SLS_RARE_5: 1.0DL + 1.0DW + 0.75LL + 1.0WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SEISMIC_2: 1.35DL + 1.75DW + 0.2LL + 0.75EL	2684.033	G2	13.500	524.216	G1	0.000	-1668.689	1382.034
SEISMIC_4: 1.0DL + 1.0DW + 0.2LL + 0.75EL	2684.033	G2	13.500	524.216	G1	0.000	-1668.689	1382.034
SLS_FREQUENT_2: 1.0DL + 1.2DW + 0.2LL + 0.6WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SEISMIC_1: 1.35DL + 1.75DW + 0.2LL + 1.5EL	2684.033	G2	13.500	524.216	G1	0.000	-1668.689	1382.034
Footpath load	275.628	G1	15.000	55.536	G1	0.000	-166.336	165.692
SLS_QP_1: 1.0DL + 1.2DW	2178.817	G1	15.000	423.606	G1	0.000	-1348.343	1347.700
ACCIDENTAL_3: 1.0DL + 1.0DW + 0.75LL	3004.637	G2	13.500	585.908	G1	0.000	-1868.117	1581.462
Envelope SLS	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995

Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
SLS_FREQUENT_4: 1.0DL + 1.0DW + 0.75LL + 0.5WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
WL Transverse	0.000	—	—	0.000	—	—	0.000	-0.000
1.0 DW	282.955	G1	15.000	54.899	G1	0.000	-175.939	175.939
1.5 EQ (b)	146.104	G1	15.000	27.761	G1	28.500	89.742	-89.742
BASIC_4: 1.0DL + 1.0DW + 1.5LL + 0.9WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
ACCIDENTAL_2: 1.0DL + 1.0DW + 0.75LL	3004.637	G2	13.500	585.908	G1	0.000	-1868.117	1581.462
WL Uplift	157.052	G1	15.000	29.841	G1	28.500	96.467	-96.467
EQ_X	0.000	—	—	0.000	—	—	-0.000	0.000
ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	3004.637	G2	13.500	585.908	G1	0.000	-1868.117	1581.462
SLS_RARE_4: 1.0DL + 1.0DW + 1.0LL + 0.6WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
BASIC_6: 1.0DL + 1.0DW + 1.15LL + 0.9WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SLS_FREQUENT_5: 1.0DL + 1.0DW + 0.2LL + 0.6WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SLS_QP_2: 1.0DL + 1.0DW	2178.817	G1	15.000	423.606	G1	0.000	-1348.343	1347.700
SLS_RARE_3: 1.0DL + 1.2DW + 0.75LL + 0.6WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
DD	1414.774	G1	15.000	274.497	G1	28.500	-879.697	879.697
EQ_Z	0.000	—	—	0.000	—	—	0.000	-0.000
Railing load	36.583	G2	15.000	7.682	G1	28.500	-20.693	21.986

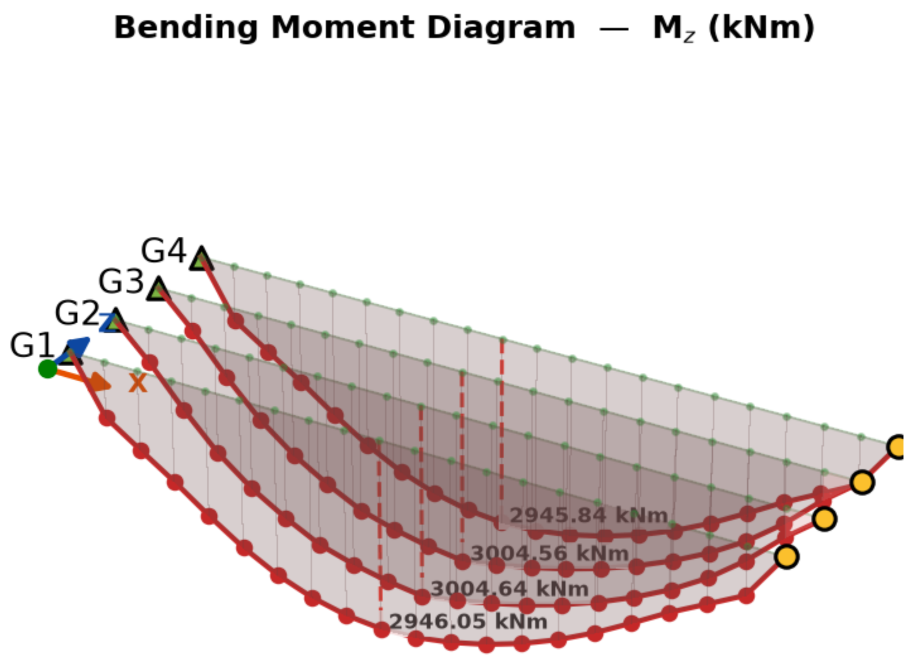
Load Case/ Comb.	Bending Moment			Shear Force			Reaction at Supports	
	Max (kNm)	Girder	Loc. (m)	Max (kN)	Girder	Loc. (m)	Left	Right
WL Longitudinal	0.000	—	—	0.000	—	—	-0.000	0.000
BASIC_1: 1.35DL + 1.75DW + 1.5LL + 0.9WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SLS_FREQUENT_6: 1.0DL + 1.0DW + 0.2LL + 0.5WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
1.0 WL	157.052	G1	15.000	29.841	G1	28.500	96.467	-96.467
BASIC_2: 1.35DL + 1.75DW + 1.15LL + 1.5WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SEISMIC_3: 1.0DL + 1.0DW + 0.2LL + 1.5EL	2684.033	G2	13.500	524.216	G1	0.000	-1668.689	1382.034
1.5 EQ (c)	487.013	G1	15.000	92.537	G1	28.500	299.141	-299.141
1.0 DL + 1.0 LL	2725.212	G2	13.500	531.008	G1	0.000	-1692.177	1405.522
SLS_FREQUENT_3: 1.0DL + 1.2DW + 0.2LL + 0.5WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
SW	205.461	G1	15.000	38.674	G1	28.500	-126.371	126.371
SLS_RARE_6: 1.0DL + 1.0DW + 0.75LL + 0.6WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
Crash barrier load	162.563	G1	15.000	31.606	G1	28.500	-94.104	99.204
1.0 DL	1895.862	G1	15.000	368.707	G1	0.000	-1172.404	1171.760
BASIC_5: 1.0DL + 1.0DW + 1.15LL + 1.5WL	2849.555	G2	13.500	556.066	G1	0.000	-1771.650	1484.995
EQ_Y	324.675	G1	15.000	61.691	G1	28.500	199.427	-199.427
Envelope ULS	3004.637	G2	13.500	585.908	G1	0.000	-1868.117	1581.462

Table 4.2: Reactions at Supports

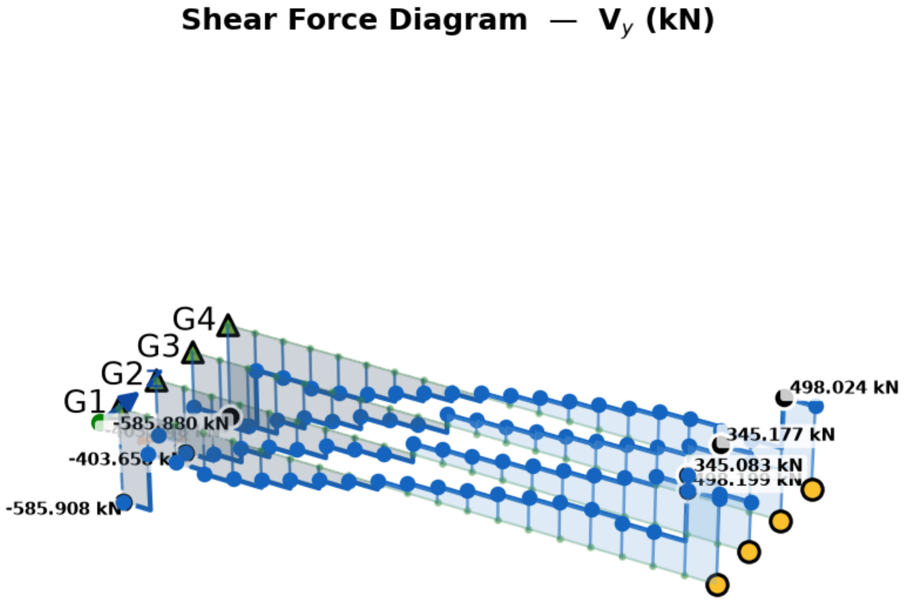
Load Case	Left Support (kN)	Right Support (kN)

Table 4.3: Deflection Summary (Live Load & Total Load)

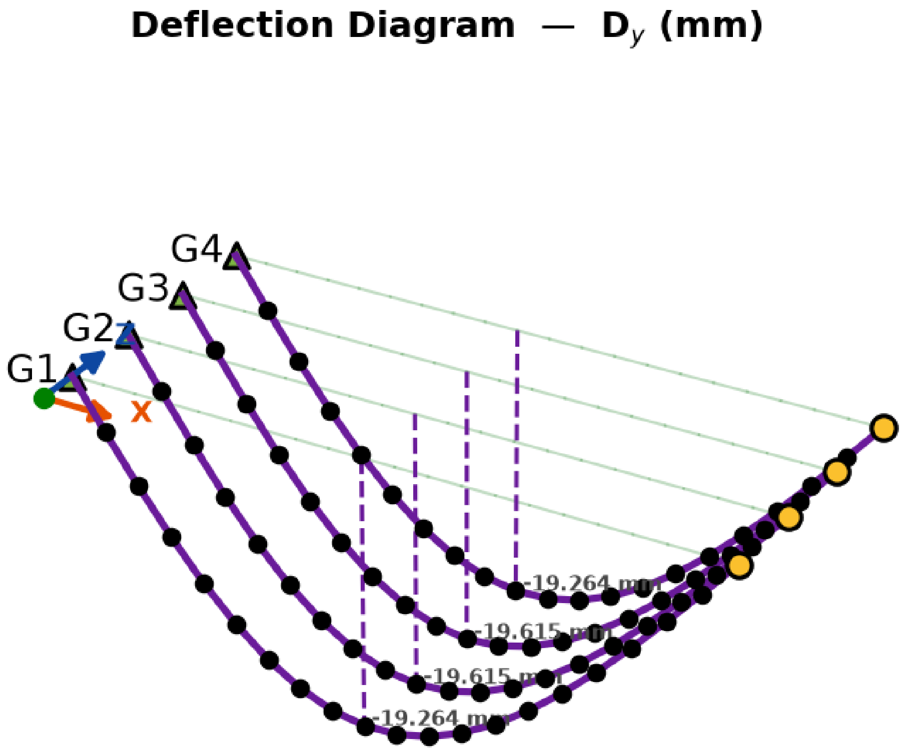
parameter	value
Deflection due to Live Load, δ_{LL}	19.615 mm
Allowable Live Load Deflection (Δ_{allow})	L/800 = 37.5 mm
Live Load Deflection Check Status	PASS
Deflection due to Total Load, δ_{total}	69.999 mm
Allowable Total Deflection (Δ_{allow})	L/600 = 50.0 mm
Total Load Deflection Check Status	FAIL



Bending Moment Envelope (Envelope ULS): Max/min BM along span. X-axis: distance from left support (m). Y-axis: Bending Moment (kN-m).



Shear Force Envelope (Envelope ULS): Max/min SF along span. X-axis: distance from left support (m). Y-axis: Shear Force (kN).



Vertical Deflection D_y (1.0 LL): Maximum deflection along span. Load Case: 1.0 LL, Combination: D_y. Nodes shown. Isometric view.

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007 and IRC 22:2014 unless stated otherwise.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Property	Value
G1	Depth, D (mm)	1670.0
	Top Flange Width, b_f (mm)	510.0
	Bottom Flange Width, b_f (mm)	510.0
	Top Flange Thickness, t_f (mm)	22.0
	Bottom Flange Thickness, t_f (mm)	22.0
	Web Thickness, t_w (mm)	10.0
	Gross Area, A (cm^2)	387.0
	Moment of Inertia, I_z (cm^4)	1881957.85
	Elastic Section Modulus, Z_{ez} (cm^3)	22538.42
	Plastic Section Modulus, Z_{pz} (cm^3)	25100.25
	Effective Slab Width, b_{eff} (mm)	3037.5
	Transformed Composite I_z (cm^4)	4515024.06
	Depth to Plastic Neutral Axis (mm)	254.5
G2	Depth, D (mm)	1670.0
	Top Flange Width, b_f (mm)	510.0
	Bottom Flange Width, b_f (mm)	510.0
	Top Flange Thickness, t_f (mm)	22.0

Girder	Property	Value
	Bottom Flange Thickness, t_f (mm)	22.0
	Web Thickness, t_w (mm)	10.0
	Gross Area, A (cm^2)	387.0
	Moment of Inertia, I_z (cm^4)	1881957.85
	Elastic Section Modulus, Z_{ez} (cm^3)	22538.42
	Plastic Section Modulus, Z_{pz} (cm^3)	25100.25
	Effective Slab Width, b_{eff} (mm)	3037.5
	Transformed Composite I_z (cm^4)	4515024.06
	Depth to Plastic Neutral Axis (mm)	254.5
G3	Depth, D (mm)	1670.0
	Top Flange Width, b_f (mm)	510.0
	Bottom Flange Width, b_f (mm)	510.0
	Top Flange Thickness, t_f (mm)	22.0
	Bottom Flange Thickness, t_f (mm)	22.0
	Web Thickness, t_w (mm)	10.0
	Gross Area, A (cm^2)	387.0
	Moment of Inertia, I_z (cm^4)	1881957.85
	Elastic Section Modulus, Z_{ez} (cm^3)	22538.42
	Plastic Section Modulus, Z_{pz} (cm^3)	25100.25
	Effective Slab Width, b_{eff} (mm)	3037.5
	Transformed Composite I_z (cm^4)	4515024.06
	Depth to Plastic Neutral Axis (mm)	254.5
G4	Depth, D (mm)	1670.0
	Top Flange Width, b_f (mm)	510.0
	Bottom Flange Width, b_f (mm)	510.0

Girder	**Property**	**Value**
	Top Flange Thickness, t_f (mm)	22.0
	Bottom Flange Thickness, t_f (mm)	22.0
	Web Thickness, t_w (mm)	10.0
	Gross Area, A (cm^2)	387.0
	Moment of Inertia, I_z (cm^4)	1881957.85
	Elastic Section Modulus, Z_{ez} (cm^3)	22538.42
	Plastic Section Modulus, Z_{pz} (cm^3)	25100.25
	Effective Slab Width, b_{eff} (mm)	3037.5
	Transformed Composite I_z (cm^4)	4515024.06
	Depth to Plastic Neutral Axis (mm)	254.5

Table 5.2: Girder Section Classification

	Element	Slenderness Ratio	Class Limit	Classification
G1	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender
G2	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender
G3	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender
G4	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender

Note: IS 800:2007 Table 2

Table 5.3: Moment Capacity Check

	Parameter	**Formula**	**Value**	**Status**
G1	Applied Moment, M_u	Governing LC (ULS)	3004.64 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11906.7 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.25	PASS
G2	Applied Moment, M_u	Governing LC (ULS)	3004.64 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11906.7 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.25	PASS
G3	Applied Moment, M_u	Governing LC (ULS)	3004.64 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11906.7 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.25	PASS
G4	Applied Moment, M_u	Governing LC (ULS)	3004.64 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11906.7 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.25	PASS

Note: IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1

Table 5.4: Shear Capacity Check

	Parameter	Formula	Value	Status
G1	Applied Shear, V_u	Governing LC (ULS)	403.66 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260.0 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.56	PASS
G2	Applied Shear, V_u	Governing LC (ULS)	403.66 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260.0 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.56	PASS
G3	Applied Shear, V_u	Governing LC (ULS)	403.66 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260.0 mm^2	—

	Parameter	Formula	Value	Status
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.56	PASS
G4	Applied Shear, V_u	Governing LC (ULS)	403.66 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260.0 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.56	PASS

Note: IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2

Table 5.5: Interaction Checks (M-V and M-N)

	Check	Condition	Value	Status
G1	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11906.7 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.26	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	0.00 + 0.25 = 0.257	0.257	PASS
G2	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11906.7 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.26	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	0.00 + 0.25 = 0.257	0.257	PASS
G3	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11906.7 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.26	PASS
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	0.00 + 0.25 = 0.257	0.257	PASS
G4	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11906.7 kN-m	—
	Interaction Check: $M_u \leq M_{dv}$	—	0.26	PASS

	Check	Condition	Value	Status
	Interaction Check: $N_u/N_{Rd} + M_u/M_{dv} \leq 1.0$	0.00 + 0.25 = 0.257	0.257	PASS

Note: IRC 22 Cl. 603.3.3.3

Table 5.6: Lateral Torsional Buckling Check – Construction Stage

	Parameter	Formula	Value	Status
G1	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	PASS
G2	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	PASS
G3	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	PASS
G4	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1.0	—

	Parameter	Formula	Value	Status
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	PASS

Note: IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2

Table 5.7: Stiffener Design Summary

Girder	Parameter	Value
G1	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	21.1
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G2	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	21.1
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G3	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	21.1
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.8

Girder	Parameter	Value
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None
G4	Shear Buckling Design Method	post_critical
	Intermediate Stiffener Thickness (mm)	21.1
	Intermediate Stiffener Spacing (mm)	
	End Panel Stiffener Thickness (mm)	1.8
	No. of End Panel Stiffeners	4
	Longitudinal Stiffeners	None

Table 5.8: End Panel Stiffener Checks

	Check	Required	Provided	Status
G1	Web Buckling Resistance	585.91 kN	168429.67 kN	PASS
	Local Crushing Resistance	585.91 kN	690.91 kN	PASS
	Bearing Capacity	585.91 kN	3181.82 kN	PASS
	Column Buckling Resistance	585.91 kN	2863.64 kN	PASS
G2	Web Buckling Resistance	585.91 kN	168429.67 kN	PASS
	Local Crushing Resistance	585.91 kN	690.91 kN	PASS
	Bearing Capacity	585.91 kN	3181.82 kN	PASS
	Column Buckling Resistance	585.91 kN	2863.64 kN	PASS
G3	Web Buckling Resistance	585.91 kN	168429.67 kN	PASS
	Local Crushing Resistance	585.91 kN	690.91 kN	PASS
	Bearing Capacity	585.91 kN	3181.82 kN	PASS
	Column Buckling Resistance	585.91 kN	2863.64 kN	PASS
G4	Web Buckling Resistance	585.91 kN	168429.67 kN	PASS
	Local Crushing Resistance	585.91 kN	690.91 kN	PASS
	Bearing Capacity	585.91 kN	3181.82 kN	PASS
	Column Buckling Resistance	585.91 kN	2863.64 kN	PASS

Note: IS 800 Cl. 8.4.2.2

Table 5.9: Serviceability – Deflection Checks

	Check	Allowable	Actual	Status
G1	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm	19.264	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm	69.588	FAIL
G2	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm	19.615	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm	69.999	FAIL
G3	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm	19.615	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm	69.996	FAIL
G4	Live Load Deflection, δ_{LL} (mm)	L/800 = 37.5 mm	19.264	PASS
	Total Load Deflection, δ_{total} (mm)	L/600 = 50.0 mm	69.580	FAIL

Note: IRC 22 Cl. 604.3.2

Table 5.10: Serviceability – Maximum Stress Limitation

	Element	Allowable Stress	Actual Stress	Status
G1	Structural Steel ($0.9\,f_y$)	315.00 MPa	104.79 MPa	PASS
G2	Structural Steel ($0.9\,f_y$)	315.00 MPa	104.79 MPa	PASS
G3	Structural Steel ($0.9\,f_y$)	315.00 MPa	104.79 MPa	PASS
G4	Structural Steel ($0.9\,f_y$)	315.00 MPa	104.79 MPa	PASS

Table 5.11: Serviceability – Fatigue Assessment

	Stress Range, $\Delta\sigma$ (MPa)	Fatigue Limit, f_{fd} (MPa)	Utilization Ratio	Status
G1	188.21 MPa	92.49 MPa	2.03	FAIL
G2	189.07 MPa	92.49 MPa	2.04	FAIL
G3	193.07 MPa	92.49 MPa	2.09	FAIL
G4	193.06 MPa	92.49 MPa	2.09	FAIL

Note: IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor μ_r applied where plate thickness ¿ 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	Controlling LC / Combination	Controlling Check	Demand	Capacity	UR	Status
G1	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	188.21 MPa	92.49 MPa	2.035	FAIL
G2	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	189.07 MPa	92.49 MPa	2.044	FAIL
G3	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	193.07 MPa	92.49 MPa	2.087	FAIL
G4	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	193.06 MPa	92.49 MPa	2.087	FAIL

Note: UR = Demand / Capacity. A value ≤ 1.0 *indicates a passing check. The controlling check is the criterion with the highest UR for each girder, with the real load case/combination that drives it.*

Table 5.13: Shear Connector Capacity

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s}, Q_{u,c})$ $Q_{u,s} = \dfrac{0.8\, f_u\, (\pi d^2/4)}{\gamma_v}$ $Q_{u,c} = \dfrac{0.29\, \alpha\, d^2 \sqrt{f_{ck}\, E_{cm}}}{\gamma_v}$	95.36 kN	IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 (ϕd, N_{sc})	25.00 kN	IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)	1314.8 mm	261.6 mm	PASS
Full Composite (SL2)	261.6 mm	261.6 mm	PASS
SLS Fatigue (SR)	479.4 mm	261.6 mm	PASS
Max Spacing Limit (IRC 22)	400.0 mm	261.6 mm	PASS

Note: IRC 22 Cl. 606.4, 606.9. Governing spacing $= \min(S_{L1}, S_{L2}, S_R)$.

Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L	145.06 kN/m	—
Transverse Shear Capacity of Slab, V_{Rd}	999.28 kN/m	—
Transverse Shear Check	$V_L/V_{Rd} = 0.15$	PASS
Min. Transverse Reinforcement, $A_{st,min}$	Required 0.72 cm^2/m, Provided 14.03 cm^2/m	PASS
Stud Diameter $\leq 2\,t_f$	$d = 20.0$ mm $\leq 2t_f = 44.0$ mm	PASS
Stud Edge Distance	Provided 195.0 mm (req. $\geq$ 25.0 mm)	PASS

Note: IRC 22 Cl. 606.6, 606.10.

5.2 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2011 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Parameter	**Value / Reference**
Effective Span of Deck Slab, l_{eff}	3038 mm (girder spacing, c/c)
Deck Thickness, t_s	250.0 mm
Clear Cover (IRC 112 Cl. 15.2)	Top 50 / Bottom 50 mm
Concrete Grade (IRC 112 Cl. 6.4)	M40 (f_{ck} = 40.0 MPa, f_{ctm} = 3.0 MPa)
Reinforcement Grade (IRC 112 Cl. 6.2)	Fe 500 (f_y = 500 MPa)
Dead Load per Unit Area, w_{DL}	6.13 kN/m² (slab self-weight)
IRC 6 Wheel Load (Class A / 70R)	83.4 kN
Tyre Contact Width (IRC 6 Annex A)	300 mm (transverse)
Impact Factor (IRC 6 Cl. 208.2)	1.100
Governing Live Load Case	Class70R(W)

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	Parameter	Formula / Reference	Value	Status
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL}\, l_{eff}^2/10$	5.66 kN-m/m	—
	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	30.25 kN-m/m	—
	Total Design BM, $M_{u,sag}$	1.35 DL + 1.50 LL	57.54 kN-m/m	—
	Effective depth, d	$t_s - c_{nom} - \phi/2$	194.0 mm	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	64.57 kN-m/m	PASS
At Support (Hogging)	Total Design BM, $M_{u,hog}$	1.35 DL + 1.50 LL (at support)	43.16 kN-m/m	—
	Required Top Steel, $A_{st,top}$	$M_u/(0.87\, f_y\, d)$	559 mm²/m	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	48.28 kN-m/m	PASS

Note: IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment (IRC 21 Cl. 305.18).

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	Formula	Value	Status
Overhang Length, l_{oh}	—	1.5187 m	—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	7.50 kN-m/m	—
Dead Load Moment	$w_{DL}\, l_{oh}^2/2$ + railing	9.31 kN-m/m	—
Live Load Moment (eccentric wheel)	Wheel load × arm	55.88 kN-m/m	—
Total Hogging Moment, $M_{u,oh}$	1.35 DL + 1.50 (LL + CB)	116.01 kN-m/m	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	124.45 kN-m/m	PASS

Note: IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	**Formula / Reference**	**Value**	**Status**
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q\,(1 + IF)\,P_w$	137.6 kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	300 × 150 mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	400 × 250 mm	—
Control Perimeter, u_1	$2(c_1 + c_2) + 4\pi d$	3738 mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed}/(u_1\,d)$	0.190 MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	0.564 MPa	—
Punching Shear Check	$v_{Ed} \leq v_{Rd,c}$	0.34	PASS

Note: Punching shear reinforcement not typically required for deck slabs with $d \geq 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	**Value / Reference**
Min. Reinforcement for Crack Control, $A_{s,min}$	303 mm²/m [IRC 112 Cl. 16.5.1]
Provided Reinforcement (bottom)	ϕ12 @ 140 mm c/c (808 mm²/m)
Max. Permissible Crack Width	0.30 mm
Calculated Crack Width, w_k (governing)	0.2117 mm
Crack Width Check	PASS

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	**Formula / Reference**	**Value**	**Status**
Design Shear per unit width, V_{Ed}	$\gamma_{DL} V_{DL} + \gamma_{LL}(1{+}IF)V_{LL}$	78.29 kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	194.0 mm	—
Size factor, k	$1 + \sqrt{200/d} \leq 2.0$	2.000	—
Long. reinforcement ratio, ρ_l	$A_{sl}/(b_w\,d) \leq 0.02$	0.0042	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c}\,b_w\,d$ (Cl. 10.3.2)	109.43 kN/m	—
One-Way Shear Check	$V_{Ed} \leq V_{Rd,c}$	0.72	PASS

Note: IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	Required / Limit	Provided	Status
Main Reinforcement — Bottom (Transverse)			
Required Area, $A_{st,req}$ (mm²/m)	755 mm²/m	808 mm²/m	PASS
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	ϕ12 @ 140 mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	303 mm²/m	808 mm²/m	PASS
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	300 mm	140 mm	PASS
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$ (mm²/m)	$\geq$ 20% of main steel	377 mm²/m	PASS
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$ (mm²/m)	559 mm²/m	595 mm²/m	PASS
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	$\geq$ 50 mm (Table 14.2)	Top 50 / Bottom 50 mm	PASS

Note: IRC 112 Cl. 16.3, IS 456 Cl. 26.5. All reinforcement provisions satisfy strength and detailing requirements.

5.3 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination (DL + LL + WL).

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
G1-G2	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G1-G2	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G3-G4	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G3-G4	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0

Note: A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.24: Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	3353	177.2	250 — PASS
	Top chord	C	3038	160.5	250 — PASS
	Bottom chord	T	3038	160.5	400 — PASS
G2-G3	Diagonal	C	3353	177.2	250 — PASS
	Top chord	C	3038	160.5	250 — PASS
	Bottom chord	T	3038	160.5	400 — PASS
G3-G4	Diagonal	C	3353	177.2	250 — PASS
	Top chord	C	3038	160.5	250 — PASS
	Bottom chord	T	3038	160.5	400 — PASS

Note: 3. Limit = 250 for compression members, 400 for tension members. $K = 1.0$ *for members with both ends pinned.*

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	40 x 40 x 3	1.5 EQ (b)	9.798	10.620	0.900	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	8.877	12.550	0.820	PASS
G2-G3	Diagonal	40 x 40 x 3	1.5 EQ (a)	6.834	45.000	0.150	PASS
	Chord	40 x 40 x 3	1.5 EQ (a)	6.191	45.000	0.140	PASS
G3-G4	Diagonal	40 x 40 x 3	1.5 EQ (b)	10.210	45.000	0.230	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	9.250	45.000	0.210	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	**Member**	**Connection**	**Section**	A_g **(mm²)**	r_{min} **(mm)**
G1-G2	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G1-G2	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G3-G4	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
G3-G4	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0

Note: A_g *= gross cross-sectional area;* r_{min} *= minimum radius of gyration.*

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit / Status
G1-G2	Diagonal	C	3353	177.2	250 — PASS
	Top chord	C	3038	160.5	250 — PASS
	Bottom chord	T	3038	160.5	400 — PASS
G2-G3	Diagonal	C	3353	177.2	250 — PASS
	Top chord	C	3038	160.5	250 — PASS
	Bottom chord	T	3038	160.5	400 — PASS
G3-G4	Diagonal	C	3353	177.2	250 — PASS
	Top chord	C	3038	160.5	250 — PASS
	Bottom chord	T	3038	160.5	400 — PASS

Note: 3. Limit = 250 for compression members, 400 for tension members. $K = 1.0$ for members with both ends pinned.

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	40 x 40 x 3	1.5 EQ (b)	9.798	10.620	0.900	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	8.877	12.550	0.820	PASS
G2-G3	Diagonal	40 x 40 x 3	1.5 EQ (a)	6.834	45.000	0.150	PASS
	Chord	40 x 40 x 3	1.5 EQ (a)	6.191	45.000	0.140	PASS
G3-G4	Diagonal	40 x 40 x 3	1.5 EQ (b)	10.210	45.000	0.230	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	9.250	45.000	0.210	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.5 Overall Design Check Summary

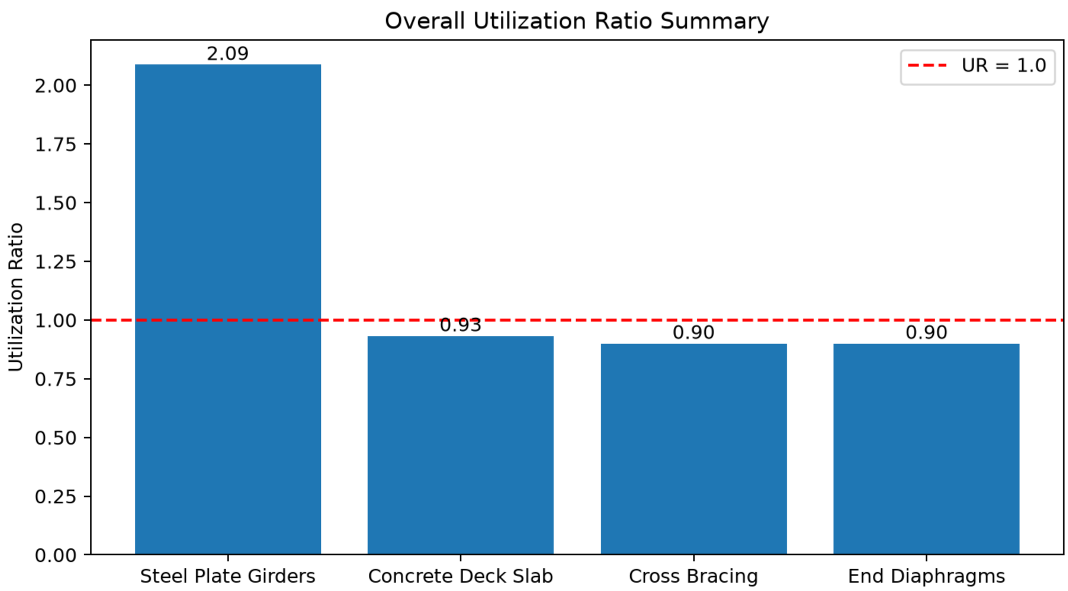


Figure 5.1: Overall Utilization Ratio Summary

Table 5.29: Overall Design Check Summary — All Members

Member / Check	Governing Load Combo	Demand	Capacity	UR
Girder — Moment	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	3004.64 kNm	11906.70 kNm	0.25
Girder — Shear	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	585.91 kN	720.21 kN	0.81
Girder — LTB (constr.)	1.0 DL	1904.90 kNm	7986.44 kNm	0.24
Girder — Deflection	1.0 DL + 1.0 LL	29.18 mm	50.00 mm	0.58
Girder — Stress	SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	111.56 MPa	315.00 MPa	0.35
Girder — Fatigue	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	193.07 MPa	92.49 MPa	2.09
Transverse Shear (slab)	—	—	—	—
Crack Width (slab)	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	0.212 mm	0.300 mm	0.71
Deck — Flexure (sagging)	Basic ULS: 1.35DL + 1.5LL	57.54 kN-m/m	64.57 kN-m/m	0.89
Deck — Flexure (hogging)	Basic ULS: 1.35DL + 1.5LL	43.16 kN-m/m	48.28 kN-m/m	0.89
Deck — Cantilever Overhang	Basic ULS: 1.35DL + 1.5LL	116.01 kN-m/m	124.45 kN-m/m	0.93
Deck — Punching Shear	Basic ULS: 1.35DL + 1.5LL	0.19 MPa	0.56 MPa	0.34
Deck — One-Way Shear	Basic ULS: 1.35DL + 1.5LL	78.29 kN/m	109.43 kN/m	0.72
Cross Bracing — Compression	1.5 EQ (b)	9.56 kN	10.620 kN	0.90

Member / Check	**Governing Load Combo**	**Demand**	**Capacity**	**UR**
Cross Bracing — Tension	1.5 EQ (b)	10.35 kN	45.000 kN	0.23
Cross Bracing — Slenderness	—	177.2	250	0.71
End Diaphragm — Moment	1.5 EQ (b)	9.56 kN	10.620 kN	0.90
End Diaphragm — Shear	1.5 EQ (b)	10.35 kN	45.000 kN	0.23

Note: UR = Demand / Capacity. All values ≤ 1.0 *indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.*

6 Drawings and Visualizations

This section presents CAD-generated views of the designed bridge and its components. All views are generated automatically by OsdagBridge using pythonOCC.

6.1 Bridge Configuration and Layout

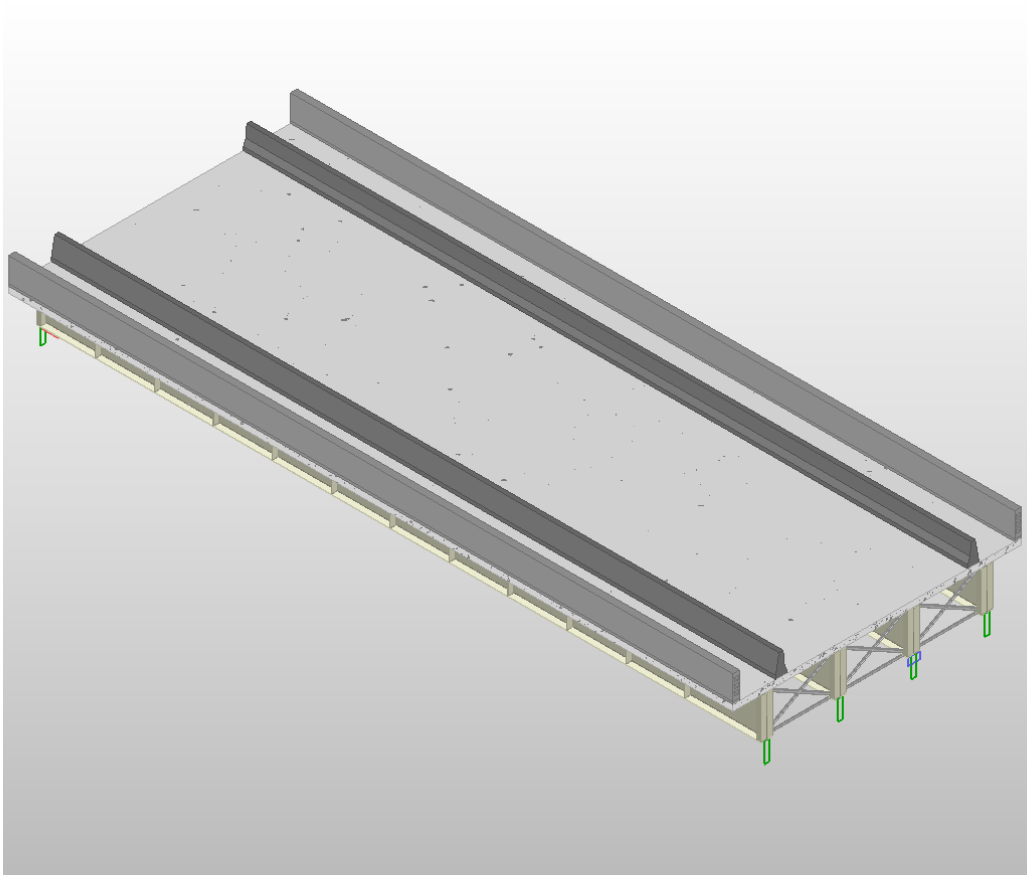


Figure 6.1: Overall 3D Bridge Superstructure

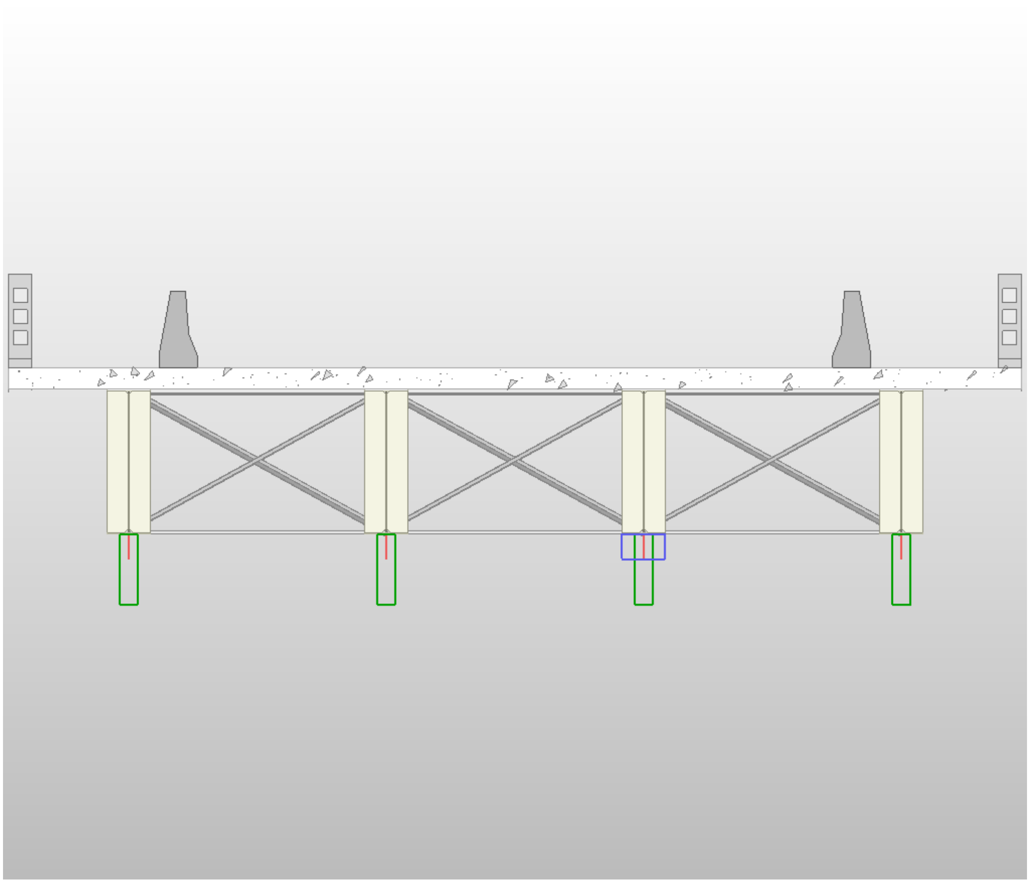


Figure 6.2: Typical Cross Section

Figure 6.3: Top View

6.2 Plate Girder — Detailed Views

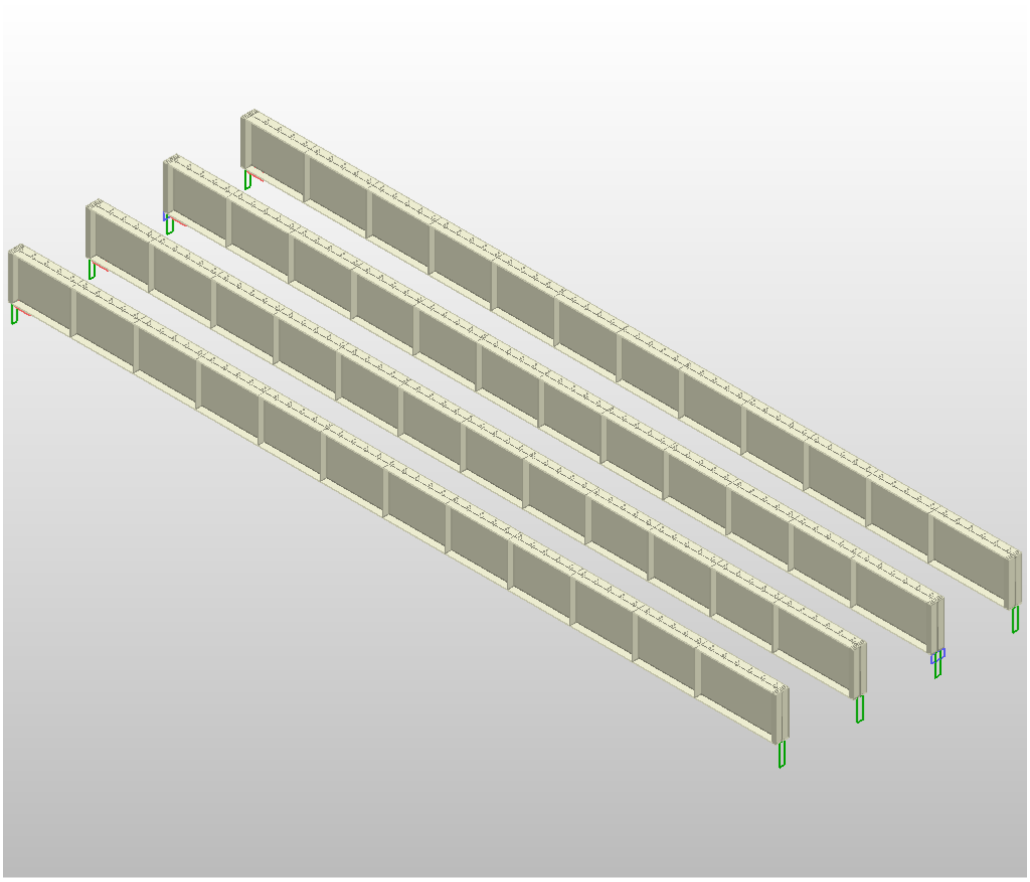


Figure 6.4: 3D View of Plate Girders

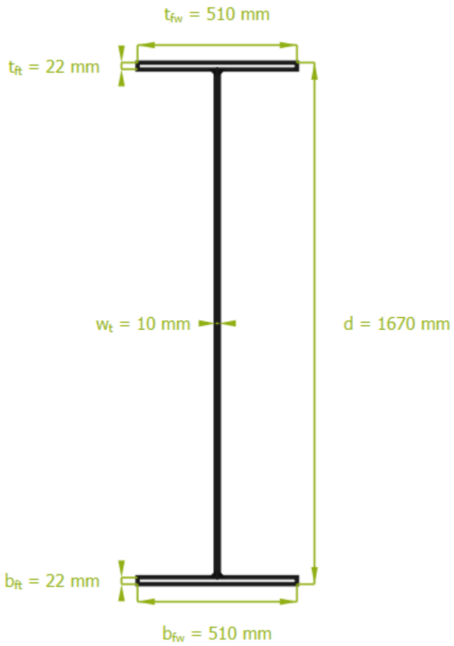


Figure 6.5: Cross Section of Plate Girder

Figure 6.6: Side View of Girder

6.3 Cross Bracing Detail

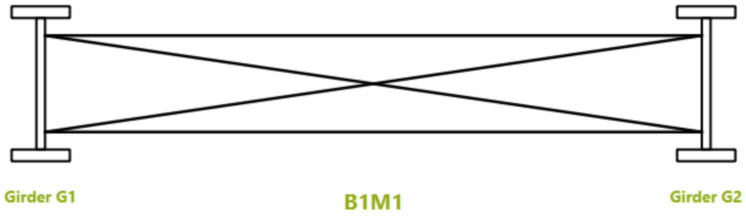


Figure 6.7: Cross Bracing Layout

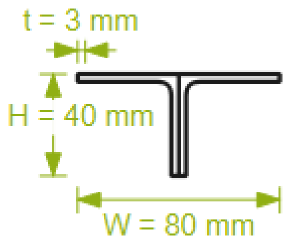


Figure 6.8: Cross Bracing – Bracing Section

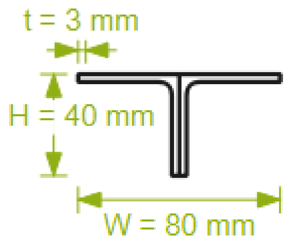


Figure 6.9: Cross Bracing – Top Chord Section

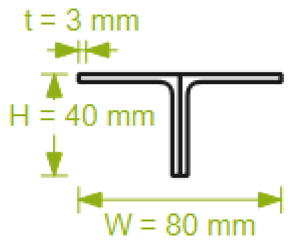


Figure 6.10: Cross Bracing – Bottom Chord Section

6.4 End Diaphragm Detail

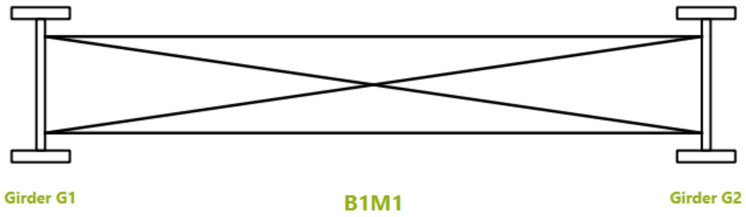


Figure 6.11: End Diaphragm Layout

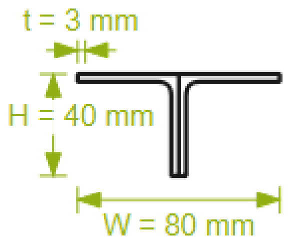


Figure 6.12: End Diaphragm – Bracing Section

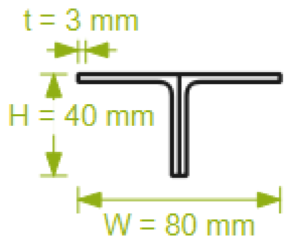


Figure 6.13: End Diaphragm – Top Chord Section

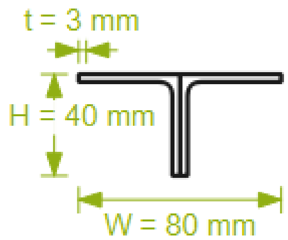


Figure 6.14: End Diaphragm – Bottom Chord Section

7 Material Take-off & Quantity Summary

Table 7.1 Bill of Materials (Steel, Concrete, and Reinforcement Quantities)

S.N.	Item Description	Volume	Quantity	Total Volume	Weight (MT)	Total Weight (MT)
1	Structural Steel (IS 2062) for Girders	$0.03492\ m^2 \times 30.00\ m = 1.04756\ m^3$	4	4.19	8.22	32.89
2(a)	Cross Bracing - Top Chord	$0.00024\ m^2 \times 3.04\ m = 0.00072\ m^3$	27	0.02	0.0057	0.15
2(b)	Cross Bracing - Bottom Chord	$0.00024\ m^2 \times 3.04\ m = 0.00072\ m^3$	27	0.02	0.0057	0.15
2(c)	Cross Bracing - Diagonal Chord	$0.00024\ m^2 \times 3.35\ m = 0.00079\ m^3$	54	0.04	0.0062	0.34
3	Concrete (M40) for Deck Slab	$12.15\ m \times 0.25\ m \times 30.00\ m = 91.12\ m^3$	1	91.12	227.81	227.81
4	Reinforcement Steel (Fe 500)	$0.046433\ m^2 \times 30.00\ m = 1.39299\ m^3$	1	1.39	10.93	10.93
5	Shear Stud Connectors	$0.000314\ m^2 \times 0.100\ m = 0.000031\ m^3$	920	0.03	0.000247	0.227
6	Crash Barrier	$0.00000\ m^2 \times 30.00\ m = 0.00001\ m^3$	2	0.00	0.00	0.00

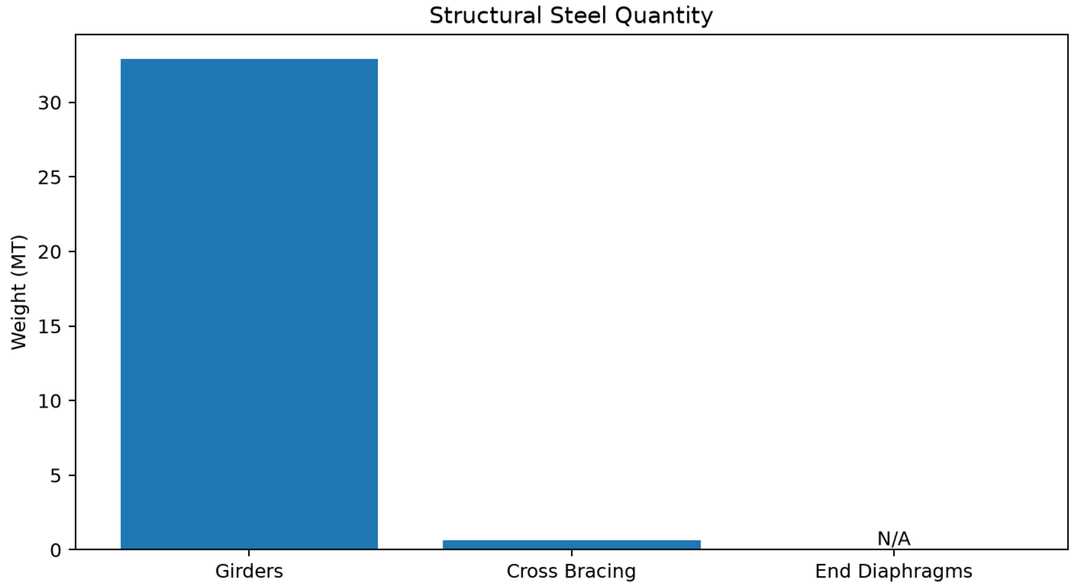


Figure 7.1: Structural Steel Quantity

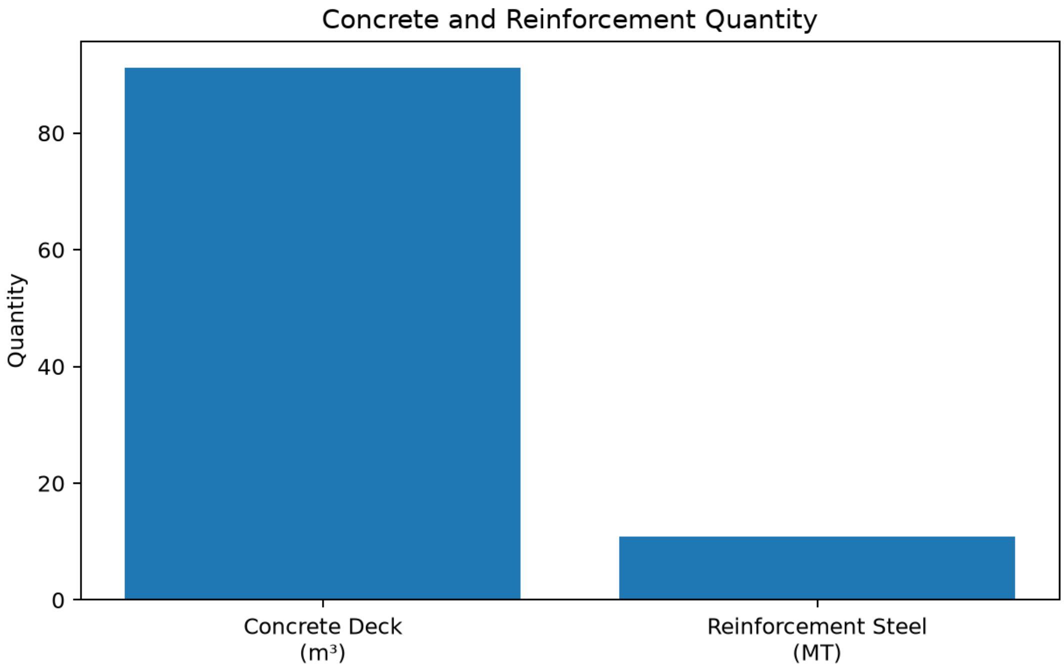


Figure 7.2: Concrete and Reinforcement Quantity

8 Standards & Assumptions

This section provides references to standards used to calibrate the OsdagBridge design modules, and notes the limitations of the current software version.

8.1 Design Standards

The following Indian Road Congress (IRC) codes and Indian Standards (IS) form the basis of all design calculations in this software.

Table 8.1: IRC Codes

Code	Year	Title / Scope
IRC 5	2015	General Features of Design - carriageway widths, kerb, footpath dimensions
IRC 6	2017	Loads and Load Combinations - dead load, live load, impact, wind, temperature, etc.
IRC 22	2015	Composite Construction (LS) - Composite section properties, ULS/SLS design, shear connectors
IRC 24	2010	Steel Road Bridges (LS) - Stiffener design, skew angle limits, diaphragm requirements
IRC 112	2020	Concrete Road Bridges - Deck slab flexure, shear, crack width, reinforcement
IRC SP 114	2018	Seismic Design of Road Bridges

Table 8.2: IS Codes

Code	Year	Scope
IS 800	2007	Steel construction - tension, compression, bending, shear, LTB, stiffeners, combined checks
IS 456	2000	Concrete - simplified stress-block for deck moment capacity
IS 1786	2008	Reinforcement steel properties
IS 1893 (Part 3)	2014	Earthquake resistant design
IS 2062	2011	Structural steel - yield and ultimate strength by grade

8.2 Analysis and Design Assumptions of This Version

Structural Analysis

- All girders are modelled as simply supported; continuous spans are not currently supported.
- A 3D grillage model (OSPGrillage) is used for load distribution. Grillage members carry composite section properties after the construction stage.
- The transverse member forces are computed using an approximate 2D frame analogy. For irregular or skewed geometries, a 3D FEM is recommended.
- Fixed bearing stiffness is modelled as $k = 1{,}000{,}000\,\mathrm{kN/m}$ (virtually rigid); free/expansion bearing as $k = 100\,\mathrm{kN/m}$.
- Construction stage sequence analysis is approximate. Detailed staged analysis should be performed for long-term deflection checks.

Composite Action

- Full shear connection is assumed at ULS with headed stud connectors designed per IRC 22:2015 Cl.606.
- Short-term composite section properties (modular ratio $n = E_s/E_{cm}$) are used for ULS checks.
- Long-term composite section properties (with creep-adjusted modular ratio) are used for SLS deflection and crack-width checks.
- Pre-composite stage: the steel girder alone resists all construction loads prior to concrete gaining strength.

Material Properties (IRC 22:2015 Annex III)

- Steel: $E_s = 200{,}000\,\mathrm{MPa}$, $G_s = 80{,}000\,\mathrm{MPa}$, $\nu = 0.30$, $\alpha = 11.7 \times 10^{-6}/°\mathrm{C}$ (grade-independent).
- Minimum structural concrete grade: M25.
- Default reinforcement grade: Fe500.

Partial Safety Factors (IRC 22:2015 Cl.601.4)

- γ_{m0} (steel yield, ULS) = 1.10
- γ_{m1} (steel ultimate, ULS) = 1.25
- Reinforcement (ULS) = 1.15
- Welds – shop: 1.25; field: 1.50
- Fatigue (γ_{mft}) = 1.35

Loading

- Dead load densities per IRC 6:2017 Cl.203: structural steel 78.5 kN/m^3, concrete 25.0 kN/m^3, bituminous wearing course 24.0 kN/m^3.
- Multi-lane live load reduction factors per IRC 6:2017 Cl.204.4 Table 6A: 1st lane = 1.0, 2nd lane = 0.8, 3rd lane onwards = 0.4.
- Impact factor (dynamic load allowance) computed from span per IRC 6:2017 Cl.208.2/208.3.
- Wind load applied as transverse, longitudinal, and vertical components per IRC 6:2017 Cl.209.3.3–209.3.5.

Serviceability Limits

- Deflection limits (IRC 22:2015 Cl.604.3.2): live load + impact $\leq L/800$; total $\leq L/600$.
- SLS stress limits: concrete $\sigma_c \leq 0.48 f_{ck}$; rebar $\sigma_s \leq 0.80 f_{yk}$; steel $f_e \leq 0.9 f_y$.
- Permissible crack width: $w_k \leq 0.3$ mm (bridge deck, exposure class XS2/XD2 per IRC 112:2020 Cl.12.3.2).

Fatigue

- Reference fatigue life: $N_{sc} = 2 \times 10^6$ cycles (IRC 22:2015 Cl.605).
- Constant stress range is assumed. A thickness correction factor μ_r is applied for plate thickness > 25 mm.

Stiffener Design

- Intermediate transverse stiffener and bearing stiffener design follows IS 800:2007 Cl.8.7.2/8.7.3 and IRC 24:2010 Cl.509.7.2/509.7.3.

8.3 Known Limitations of This Version

- Substructure (piers, pile caps, foundations) and bearing design are not included.
- Splice connection design is not implemented.
- Skew angle > 15 degrees requires independent manual analysis (IRC 24 Cl. 504.8).
- Construction stage sequence analysis is approximate; detailed staged analysis should be performed for long-term deflection checks.
- The grillage analysis assumes simply supported boundary conditions; continuous spans are not currently supported.

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*

2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*

3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*

4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*

5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*

6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*

7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*

8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*

9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.

10. Steel-INSDAG Teaching Resource Materials. `https://www.steel-insdag.org`